

1. Vector calculus

In this problem we recall a few more standard identities of vector calculus which we will frequently use in electrodynamics.

The components of a three-dimensional vector product $\vec{a} \times \vec{b}$ are given by

$$(\vec{a} \times \vec{b})_i = \sum_{j,k=1}^3 \varepsilon_{ijk} a_j b_k. \quad (1.1)$$

Here, ε_{ijk} is the totally anti-symmetric tensor in $\mathbb{R}^3$ with $\varepsilon_{123} = +1$.

Recall that you have already shown that

$$\sum_{i=1}^3 \varepsilon_{ijk} \varepsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl} \quad \text{and} \quad \sum_{i,j=1}^3 \frac{1}{2} \varepsilon_{ijk} \varepsilon_{ijl} = \delta_{kl}. \quad (1.2)$$

Prove the following identities for arbitrary scalar fields F and vector fields $\vec{A}$, $\vec{B}$:

a) $\vec{\partial} \times (F \vec{A}) = (\vec{\partial} F) \times \vec{A} + F \vec{\partial} \times \vec{A}$

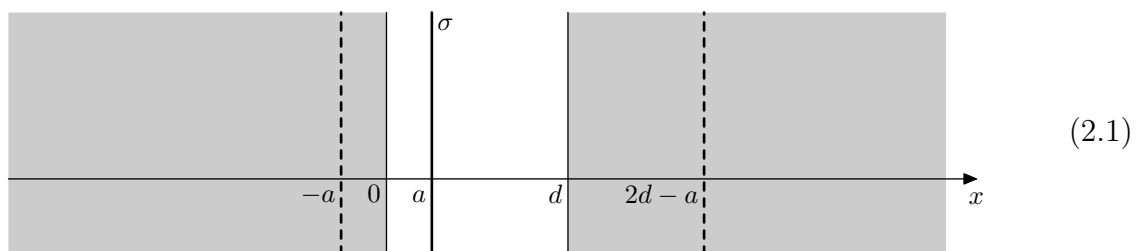
b) $\vec{\partial}(\vec{A} \cdot \vec{B}) = (\vec{A} \cdot \vec{\partial}) \vec{B} + (\vec{B} \cdot \vec{\partial}) \vec{A} + \vec{A} \times (\vec{\partial} \times \vec{B}) + \vec{B} \times (\vec{\partial} \times \vec{A})$

c) $\vec{\partial} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\partial} \times \vec{A}) - \vec{A} \cdot (\vec{\partial} \times \vec{B})$

2. Iterated images

Green functions are a fundamental tool for solving problems in electrostatics and, more generally, differential equations.

- a) What is the differential equation that defines a Green function in electrostatics? What is its physical interpretation?



Consider the region of space between two conducting, grounded, infinitely extended parallel plates that lie at $x = 0$ and $x = d$. Suppose that a third charged plate with surface charge density σ is placed at $x = a$ where $0 < a < d$. Incidentally, since the system is invariant under translations orthogonal to the x -axis, finding the potential generated by the third plate corresponds to computing a one-dimensional Green function.

- b) Is it possible to find a distribution of image charges outside of the considered physical region of space that yields the Green function for the given charge distribution and boundary conditions?

3. Multipole expansion

We consider the multipole expansion of the electrostatic potential in cartesian coordinates.

- a) What is the physical context in which we typically talk about multipole moments?
What is the defining expansion?

- b) What is the form of the electrostatic potential of an ideal dipole?

- c) Consider a charge configuration consisting of two point charges q at position y and $-q$ at position $-y$. Find the monopole, dipole and quadrupole moments of this charge distribution. What can you tell about the higher moments?

4. Sphercial harmonics

- a) Combine the spherical harmonics $Y_{1,m}$ into a vector $\vec{Y}_1$ which transforms canonically under rotations of space.

Hint: $Y_{1,0} = \sqrt{3} \cos \vartheta$, $Y_{1,\pm 1} = \mp \sqrt{3/2} \sin \vartheta e^{\pm i\varphi}$.

- b) Consider homogeneous quadratic polynomials in the coordinates x, y, z . Among these, find a basis of eigenfunctions for the operator $\vec{x}^2 \Delta$. Relate the corresponding eigenvalues to the eigenvalues of the operator $\vec{L}^2$ for the spherical harmonics.

Hint: $\Delta = \partial_r^2 + 2r^{-1} \partial_r - r^{-2} \vec{L}^2$.

5. Motion of a current-carrying wire

Consider three straight, infinitely long, parallel and coplanar wires (with zero radius), each carrying a current I in the same direction. Let the distance between each of the outer wires to the one in the middle be d .

- a) Argue that the two zeros in the magnetic field have to be located in the same plane as the wires. Then calculate these locations.

- b) Sketch the magnetic field line pattern.

- c) The middle wire is now displaced by a very small distance x ($x \ll d$) out of the plane and perpendicular to the axis on which the wires sit. It is subsequently free to move while the other two wires are held fixed at their original position. Describe qualitatively the motion of the middle wire (no calculation!).

6. Gauge transformations

- a) Consider two vector potentials $\vec{A}_1, \vec{A}_2$ that generate the same magnetic field $\vec{B} = \vec{\partial} \times \vec{A}_1 = \vec{\partial} \times \vec{A}_2$. Show that they are locally related by a gauge transformation $\vec{\partial} \Lambda$.

In what follows, we consider two hypothetical vector potentials in spherical coordinates

$$\vec{A}_1 = \kappa \frac{1 - \cos \vartheta}{r \sin \vartheta} \vec{e}_\varphi \quad \text{and} \quad \vec{A}_2 = -\kappa \frac{1 + \cos \vartheta}{r \sin \vartheta} \vec{e}_\varphi. \quad (6.1)$$

- b) Show that the two vector potentials are equivalent by finding a suitable gauge transformation $\Lambda(x)$. What is noteworthy about the potentials or the gauge transformation?
Hint: Recall the gradient formula $\vec{\partial} \Lambda = \vec{e}_r \partial_r \Lambda + (\vec{e}_\vartheta / r) \partial_\vartheta \Lambda + (\vec{e}_\varphi / r \sin \vartheta) \partial_\varphi \Lambda$.

- c) The magnetic field evaluates to $\vec{B} = (\kappa/r^2) \vec{e}_r$. What can you say about this field configuration?

- a) Determine the inductance L_{tot} of two solenoids in parallel with inductances L_1 and L_2 . Neglect any inductive couplings.
- b) Determine the inductance L_{tot} of two solenoids in series with inductances L_1 and L_2 . Neglect any inductive couplings.
- c) A solenoid with inductance L is connected to a voltage source U_0 via a resistance R . Determine the differential equation for the current $I(t)$ and solve it for $I(0) = 0$.

8. Four-vectors and Poincaré transformations

- a) Consider the Poincaré transformation $x'^\mu = (\Lambda^{-1})^\mu{}_\nu x^\nu - b^\mu$.

Describe how the following objects behave under this transformation (covariant, contravariant, invariant, tensorial with rank $\dots$, other):

$$x^\mu, \quad t, \quad \vec{x}, \quad \partial^2, \quad \partial_\mu, \quad \partial^\mu, \quad dx^\mu, \quad d = dx^\mu \partial_\mu, \quad dx^4, \quad \eta^{\mu\nu}, \quad \varepsilon_{\mu\nu\rho\sigma}, \quad \partial^\nu F_{\mu\nu}. \quad (8.1)$$

For those objects which have a non-standard transformation, can you say under which subgroup of the Poincaré group they follow a standard rule (invariant, covariant, $\dots$)?

- b) Write down the transformation rules for the potential $A_\mu(x)$ and the field strength $F_{\mu\nu}(x)$. Is there any difference between the transformation of $F_{\mu\nu}$ and its dual $\tilde{F}_{\mu\nu}$?

- c) Show that the field strength $F_{\mu\nu}$ constructed from the potential A_μ satisfies the homogeneous Maxwell equation. Furthermore, derive the conservation of the current density J_μ from the inhomogeneous Maxwell equation.

9. Polarisers

Consider a monochromatic wave polarised in the x -direction, propagating along the z -axis. Its electric field is of the form $\vec{E} = \text{Re}[\psi(z, t)]\vec{e}_x$ where $\psi(z, t) = A e^{ikz - i\omega t}$ and its intensity I_0 is proportional to $|\psi(z, t)|^2 = |A|^2$. The wave passes a linear polariser whose transmission axis is tilted by an angle β with respect to the x -axis.

a) What is the intensity I of the transmitted wave?

b) In front of the polariser one now places a quarter wave plate whose optical axis is tilted by an angle α with respect to the x -axis: The phase of the electric field component parallel to the optical axis is shifted by $\pi/2$ while the perpendicular component remains unchanged. The plate therefore produces elliptically polarised light which then passes the linear polariser. Show that the intensity I of the outgoing wave behind the linear polariser is given by $I = I_0 \{ [\cos \alpha \cos(\beta - \alpha)]^2 + [\sin \alpha \sin(\beta - \alpha)]^2 \}$.

c) Show that for $\alpha = \pi/4$ the intensity I of the transmitted wave becomes independent of β . Why? What is the value for I in this case?

10. Time evolution of electromagnetic waves

Consider an electromagnetic wave with wave number $\vec{k} = k\vec{e}_z$, amplitude A and initial conditions $E_0(\vec{x})$ and $B_0(\vec{x})$ at time $t = 0$ given by

$$\vec{E}_0(\vec{x}) = A \cos(kz) \vec{e}_x, \quad \vec{B}_0(\vec{x}) = -\frac{A}{c} \cos(kz) \vec{e}_y. \quad (10.1)$$

We want to construct the time evolution of this wave.

- a) Based on the free wave equation $\square \vec{E} = \square \vec{B} = 0$ and the particular spatial dependency of $\vec{E}_0$ and $\vec{B}_0$, what are the admissible (four complex) basis functions for wave solutions?
- b) Make an ansatz for E_x and B_y as linear combinations of these basis functions.
- c) Impose the Maxwell equations to constrain the coefficients.
- d) Restrict the coefficients further such that the initial conditions are satisfied.
- e) Compose the resulting expressions into real functions for $\vec{E}$ and $\vec{B}$.

11. Radiation of a circular orbit

- a) A (non-relativistic) particle with charge q rotates on a circular orbit in the x,y -plane of radius r and angular velocity ω . The path of the particle reads in complex form

$$\vec{x}(t) = \text{Re}(\vec{x}_0 e^{-i\omega t}), \quad \vec{x}_0 = r(\vec{e}_x + i\vec{e}_y). \quad (11.1)$$

Compute its dipole moments $\vec{P} = q\vec{x}$ and $\vec{M} = \frac{1}{2}q\vec{x} \times \dot{\vec{x}}$ and argue why the magnetic dipole moment will not contribute to radiation.

- b) The radiation due to the dipole moment is

$$\frac{d^2 P}{d^2 \Omega} = \frac{c^3 k^4 \mu_0}{32\pi^2} \langle \|\vec{P}\|^2 - |\vec{P} \cdot \vec{n}|^2 \rangle = \frac{\omega^4 q^2 r^2 \mu_0}{32\pi^2 c} (1 + \cos^2 \vartheta). \quad (11.2)$$

Discuss the radiation profile: What is the ratio of intensities along the x,y,z -axes? How does it arise? Can you say something about the polarisation of the radiation?

- c) What can you say about radiation from the electric quadrupole moment $R(t) = q(3\vec{x}\vec{x}^\top - \vec{x}^2)$?

- d) What will change qualitatively if the particle rotates at a relativistic velocity?

12. Photography and Fata Morgana

- a) You would like to take a photograph of an object behind a window, but a light source on your side of the window causes a distracting reflection on the glass pane. Fortunately, your camera has a polariser. What do you do in order to optimise the photo?

Hint: The refractive index of air is $n = 1$, the one of the glass is $n_G \approx 1.5$.

- b) On warm summer days, the phenomenon of mirage (“Fata Morgana”) can sometimes be observed above surfaces which have been heated by the sun. The underlying reason for this is a temperature dependent refractive index. The air layers above the ground become colder with increasing height, the density and thus the refractive index increases.

Explain qualitatively (no calculation!) why incoming light may not reach the ground but gets reflected in the heated air layers which generates the impression of a water surface above the ground.